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TRIAD: Benchmarking Omni-Modal Ambiguity in Multimodal Large Language Models

A New Benchmark Reveals That Even the Best Omni-Modal Systems Trail Humans by Over 20 Points When All Three Modalities Are Needed to Resolve a Scene

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Multimodal language models that process text, images, and audio are routinely evaluated on benchmarks where a single dominant modality carries the answer—leaving intact an unmeasured gap between claimed omni-modal capability and the genuine ability to fuse all three modalities simultaneously. TRIAD closes this gap with 327 hand-crafted queries spanning 86 scene groups and 18 ambiguity categories, each designed so that the complete text–image–audio triplet yields a unique answer while any single-modality subset leaves two or more options open. Across 14 state-of-the-art omni-modal systems, the best model trails humans by over 20 group-accuracy points, and removing answer options drives model group accuracy to single digits.

AT A GLANCE

Problem: Existing benchmarks do not require models to integrate all three modalities; a single modality usually suffices to answer correctly.

Why it matters: Omni-modal systems are deployed as universal assistants; a model that silently ignores audio will fail in safety-critical settings where acoustic context is essential.

Method: 327 text–image–audio triplets where the correct answer requires all three modalities, evaluated on item and scene- group accuracy with and without answer options.

Result: Best model trails humans by >20 points on group accuracy; removing answer options drops model group accuracy to single digits.

Evidence: 14 state-of-the-art omni-modal systems on 86 scene groups covering 18 ambiguity categories.

The Big Picture

The ambition of omni-modal language models—processing text, image, and audio within a single system—promises agents that perceive the world as richly as humans do. Yet evaluating this ambition has relied on benchmarks that are functionally bimodal: image-question answering where text alone is sufficient, or audio tasks where the image adds marginal signal. True omni-modal integration requires all three modalities to be jointly necessary.

TRIAD operationalizes this requirement through the principle of *modal indispensability*: for each query, every modality is necessary and no proper subset is sufficient. The dataset captures everyday scenes where text, image, and audio together resolve what neither pair can—e.g., hearing an alarm while seeing a clock while reading a date.

Why Existing Approaches Fall Short

Item-level evaluation is misleading. A model that achieves 70% accuracy on individual items may still fail every scene group, because group accuracy requires consistent disambiguation across all items sharing a scene. TRIAD reports both metrics; the gap between them exposes unreliable partial reasoning.

Answer options inflate performance. Multiple-choice formats let models exploit statistical regularities in the option set (e.g., always picking the longest option). TRIAD evaluates both with and without answer options; performance drops to single digits in the open-ended condition, confirming option exploitation.

Audio cues are underweighted. Models trained on predominantly text–vision data default to text–image reasoning even when the disambiguating signal is acoustic. TRIAD's audio-indispensable categories expose this systematic underweighting.

Core Mechanism

TRIAD's benchmark construction follows three stages:

1. **Scene collection:** Authors collect or synthesize naturalistic scenes representable in all three modalities with complementary but partially overlapping information.
2. **Ambiguity verification:** Each triplet is manually reviewed to confirm that any two-modality combination leaves at least two consistent answer options.
3. **Query authoring:** Questions are written to be answerable only with the full triplet; distractors are crafted to be plausible given incomplete modality access.

Scene groups (86 in total) share a common real-world context and each contain 3–5 queries, allowing the group accuracy metric to test whether disambiguation general-

izes within a scenario.

Technical Formulation

Let $\mathcal{T} = (t, v, a)$ denote a triplet of text t, image v, and audio a, and let $\mathcal{A} = \{a_1, \ldots, a_K\}$ be the answer set. A triplet is modally indispensable if:

$$|\arg\max_k P(a_k \mid \mathcal{T})| = 1 \quad \text{but} \quad |\arg\max_k P(a_k \mid \mathcal{T}\backslash\{m\})| > 1$$

for every modality $m \in \{t, v, a\}$. This condition defines the 327 items in TRIAD; it is verified by human annotators, not automatically.

Group accuracy for scene group g is:

$$\text{GroupAcc}(g) = \mathbf{1}[\forall q \in g,\ \hat{a}(q) = a^*(q)]$$

where $\hat{a}(q)$ is the model's predicted answer and $a^*(q)$ the ground-truth. Mean group accuracy across all 86 groups is the primary metric.

Learning or Inference Procedure

TRIAD is a static evaluation benchmark; no training is involved. Models are prompted with the triplet in their native input format (text + image + audio), asked to output either a multiple-choice letter or a free-form answer, and scored automatically. Human performance is estimated from a held-out annotator pool distinct from the construction team.

What the Guarantee Says

The modal-indispensability design guarantee is:

- **Soundness:** A model that answers correctly must have accessed all three modalities (no single-modality shortcut exists by construction).
- **Completeness:** Group accuracy measures consistent disambiguation within a scene, ruling out lucky individual item guesses.

These are guarantees about the benchmark's diagnostic validity, not about any particular model's capability.

Experimental Findings

- **Group accuracy gap:** The best of 14 evaluated omni-modal systems achieves group accuracy >20 points below human performance.
- **Item vs. group accuracy:** Models with high item accuracy (e.g., >65%) still fail most scene groups, confirming inconsistency.
- **Open-ended collapse:** Removing answer options drops average model group accuracy to single digits; human group accuracy declines only modestly (from ∼85% to ∼70%).
- **Audio underuse:** On audio-indispensable items, models score near chance even in the multiple-choice condition, confirming that acoustic reasoning is the primary failure mode.

Ablations and Interpretation

Modality ablation: Providing models with only text+image, text+audio, or image+audio gives performance close to random, confirming benchmark validity.

Model family comparison: Closed-source frontier models outperform open-source models by a larger margin on TRIAD than on existing benchmarks, suggesting that scale and instruction tuning improve omni-modal integration but do not close the gap.

Category breakdown: The hardest ambiguity categories involve temporal audio events (e.g., distinguishing two events from their sounds alone) and spatial sound-source localization, pointing toward specific architectural improvements.

Reference

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